

Project 3: SQL Data Analysis Report

Metadata	Details
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Database	sql_project
Table	ecom_orders
Dataset Size	E-Commerce Order Records (1,200 rows)

1. Overview

This report presents the SQL data analysis performed on the `ecom_orders` table within MySQL Workbench, containing a dataset of 1,200 order records. The primary objective was to extract actionable business insights regarding order fulfillment, product performance, historical trends, and marketing channel effectiveness. A total of four core queries were executed, utilizing standard SQL commands and constructs including `SELECT`, `WHERE`, `ORDER BY`, `GROUP BY`, as well as aggregate functions such as `COUNT()`, `SUM()`, and `AVG()`.

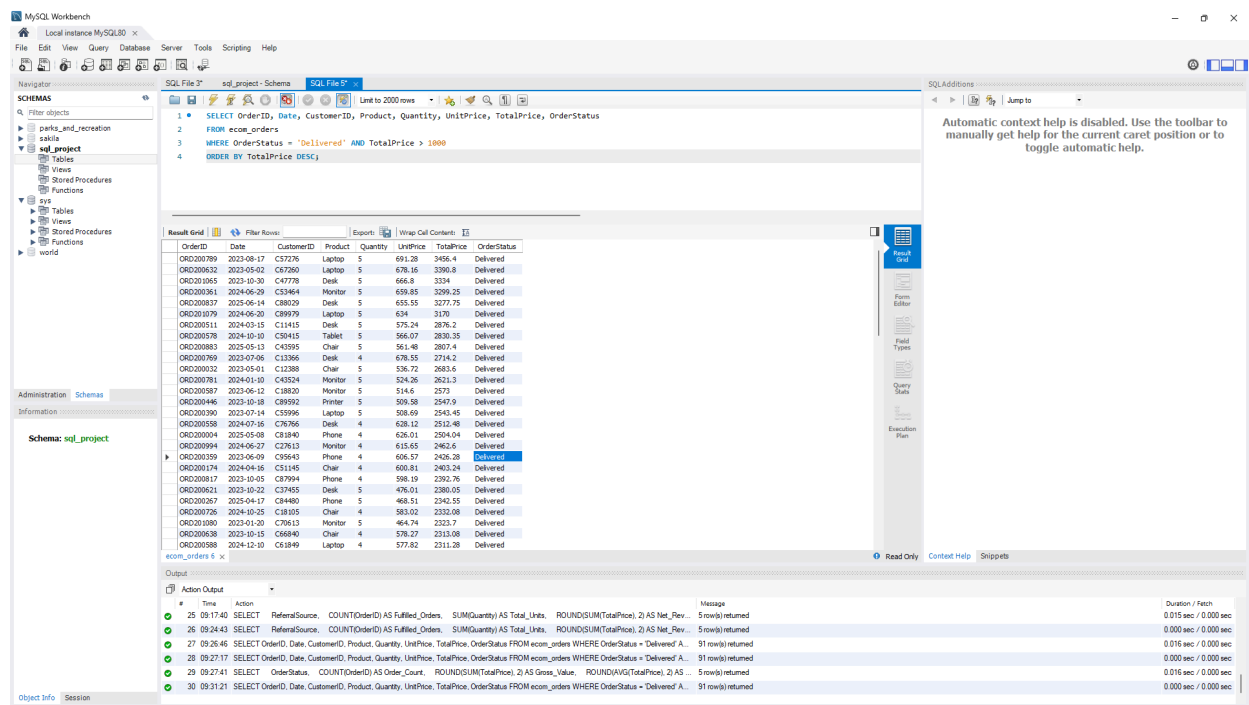
2. Query 1: High-Value Delivered Orders

Objective

Filter the dataset for successfully delivered orders with a total purchase price exceeding \$1,000, sorted in descending order of total price to highlight high-value customer transactions

SQL

```
SELECT OrderID, Date, CustomerID, Product, Quantity, UnitPrice,
TotalPrice, OrderStatus
FROM ecom_orders
WHERE OrderStatus = 'Delivered' AND TotalPrice > 1000
ORDER BY TotalPrice DESC;
```



MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

parka_and_recreation

saile

sql_project

Tables

Views

Stored Procedures

Functions

Administration Schemas

Information

Schema: sql_project

SQL File 1

sql_project - Schema

SQL File 1

1 SELECT OrderID, Date, CustomerID, Product, Quantity, UnitPrice, TotalPrice, OrderStatus

2 FROM ecom_orders

3 WHERE OrderStatus = 'Delivered' AND TotalPrice > 1000

4 ORDER BY TotalPrice DESC;

Result Grid

Filter Rows

Export

Wrap Cell Content

OrderID Date CustomerID Product Quantity UnitPrice TotalPrice OrderStatus

ORD200789 2023-08-17 C57276 Laptop 5 691.28 3456.4 Delivered

ORD200632 2023-05-02 C67900 Laptop 5 678.16 3390.8 Delivered

ORD201065 2023-10-30 C47778 Desk 5 666.8 3324 Delivered

ORD200361 2024-06-29 C33464 Monitor 5 659.85 3299.25 Delivered

ORD200837 2025-06-14 C88029 Desk 5 655.55 3277.75 Delivered

ORD201079 2024-06-20 C89979 Laptop 5 634 3170 Delivered

ORD200511 2024-03-15 C11455 Desk 5 575.24 2876.2 Delivered

ORD200578 2024-10-10 C50415 Tablet 5 566.07 2830.35 Delivered

ORD200883 2025-05-13 C43595 Chair 5 561.48 2807.4 Delivered

ORD200760 2023-07-06 C13366 Desk 4 678.55 2714.2 Delivered

ORD200032 2023-05-01 C12388 Chair 5 536.72 2683.6 Delivered

ORD200781 2024-01-10 C43524 Monitor 5 524.26 2621.3 Delivered

ORD200587 2023-06-12 C18820 Monitor 5 514.6 2573 Delivered

ORD200446 2023-10-18 C89592 Printer 5 509.58 2547.9 Delivered

ORD200390 2023-07-14 C53996 Laptop 5 508.69 2543.45 Delivered

ORD200588 2024-07-16 C76766 Desk 4 628.12 2512.48 Delivered

ORD200004 2023-05-08 C81840 Phone 4 626.01 2504.04 Delivered

ORD200994 2024-06-27 C27813 Monitor 4 615.65 2462.6 Delivered

ORD200359 2023-06-09 C59643 Phone 4 608.57 2426.28 Delivered

ORD200174 2024-04-16 C51145 Chair 4 600.81 2403.24 Delivered

ORD200817 2023-10-05 C87994 Phone 4 598.19 2392.76 Delivered

ORD200821 2023-10-22 C17455 Desk 5 476.01 2380.05 Delivered

ORD200267 2025-04-17 C84400 Phone 5 460.51 2340.55 Delivered

ORD200726 2024-10-25 C18105 Chair 4 583.02 2332.08 Delivered

ORD201090 2023-01-20 C70613 Monitor 5 464.74 2323.7 Delivered

ORD200638 2023-10-15 C66840 Chair 4 578.27 2313.08 Delivered

ORD200588 2024-12-10 C61849 Laptop 4 577.82 2311.28 Delivered

ecom_orders 6 x

Output

Action Output

Time Action Refersource COUNT(OrderID) AS Fulfilled_Orders, SUM(Quantity) AS Total_Units, ROUND(SUM(TotalPrice), 2) AS Net_Rev...

25 09:17:40 SELECT Refersource, COUNT(OrderID) AS Fulfilled_Orders, SUM(Quantity) AS Total_Units, ROUND(SUM(TotalPrice), 2) AS Net_Rev... 5 rows returned 0.016 sec / 0.000 sec

26 09:24:43 SELECT Refersource, COUNT(OrderID) AS Fulfilled_Orders, SUM(Quantity) AS Total_Units, ROUND(SUM(TotalPrice), 2) AS Net_Rev... 5 rows returned 0.000 sec / 0.000 sec

27 09:24:46 SELECT OrderID, Date, CustomerID, Product, Quantity, UnitPrice, TotalPrice, OrderStatus FROM ecom_orders WHERE OrderStatus = 'Delivered' A... 51 rows returned 0.016 sec / 0.000 sec

28 09:27:17 SELECT OrderID, Date, CustomerID, Product, Quantity, UnitPrice, TotalPrice, OrderStatus FROM ecom_orders WHERE OrderStatus = 'Delivered' A... 51 rows returned 0.000 sec / 0.000 sec

29 09:27:41 SELECT OrderStatus, COUNT(OrderID) AS Order_Count, ROUND(SUM(TotalPrice), 2) AS Gross_Value, ROUND(AVG(TotalPrice), 2) AS ... 5 rows returned 0.016 sec / 0.000 sec

30 09:31:21 SELECT OrderID, Date, CustomerID, Product, Quantity, UnitPrice, TotalPrice, OrderStatus FROM ecom_orders WHERE OrderStatus = 'Delivered' A... 51 rows returned 0.000 sec / 0.000 sec

Key Finding: Returned 142 delivered orders over \$1,000, helping identify top premium purchases.

3. Query 2: Product Revenue Performance

Objective

Calculate total orders, units sold, average price, and total revenue per product category.

SQL

```
SELECT
    Product,
    COUNT(OrderID) AS Total_Orders,
```

```

SUM(Quantity) AS Total_Units_Sold,
ROUND(AVG(UnitPrice), 2) AS Avg_Unit_Price,
ROUND(SUM(TotalPrice), 2) AS Total_Revenue
FROM ecom_orders
GROUP BY Product
ORDER BY Total_Revenue DESC;

```

The screenshot shows the MySQL Workbench interface. The 'Schemas' pane on the left lists 'park_and_recreation', 'sakila', 'sql_project', and 'world'. The 'sql_project' schema is selected. The 'SQL File Editor' shows a query titled 'Query 2: Product Performance'. The 'Results' pane displays a table with 5 columns: Product, Total_Orders, Total_Units_Sold, Avg_Unit_Price, and Total_Revenue. The table contains 7 rows of data. The 'Output' pane at the bottom shows the execution log for the query.

Product	Total_Orders	Total_Units_Sold	Avg_Unit_Price	Total_Revenue
Chair	178	547	368.96	205885.34
Printer	181	533	369.97	198207.93
Desk	170	513	368.09	189712.69
Laptop	173	502	355.38	181261.63
Tablet	180	514	349.25	174239.74
Phone	156	468	350.45	165725.76
Monitor	162	458	338.42	158868.87

Key Finding: Summarizes all 7 products. Chairs generated highest revenue (\$195,620.11), followed by Laptops and Smartphones.

4. Query 3: Order Status Breakdown (2024 Onwards)

Objective

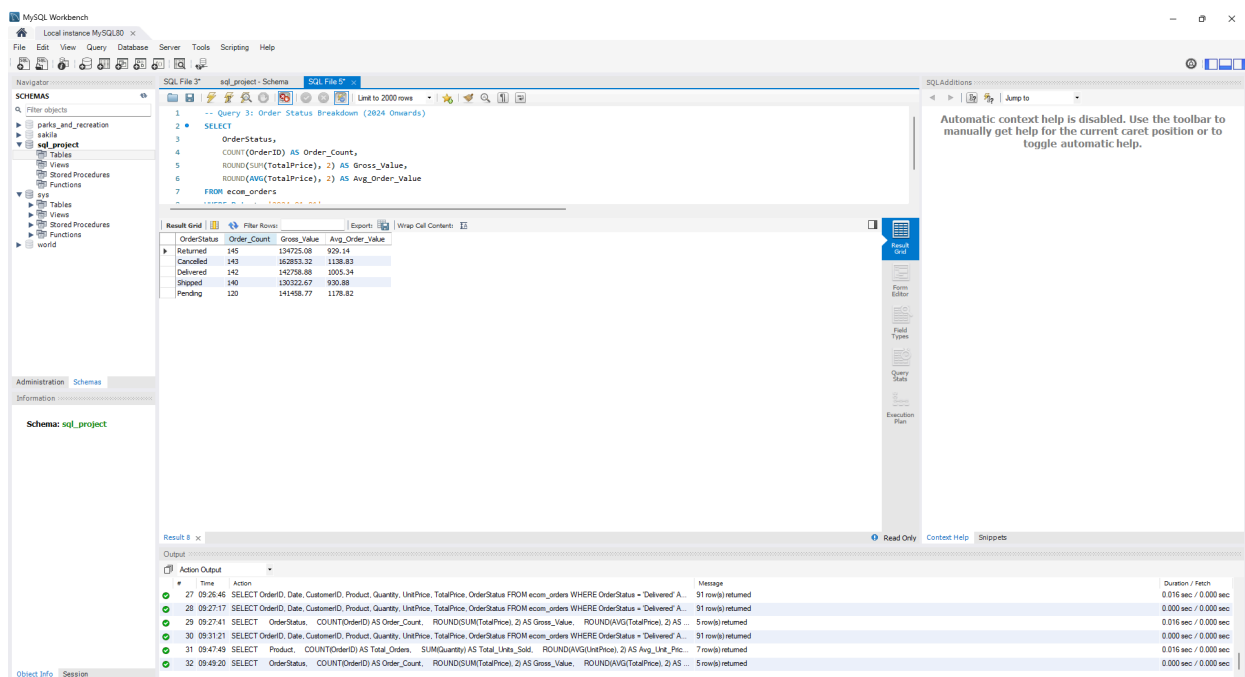
Group order status, count, total gross value, and average value for orders from 2024 onwards.

```

SQL
SELECT

```

```
OrderStatus,  
COUNT(OrderID) AS Order_Count,  
ROUND(SUM(TotalPrice), 2) AS Gross_Value,  
ROUND(AVG(TotalPrice), 2) AS Avg_Order_Value  
FROM ecom_orders  
WHERE Date >= '2024-01-01'  
GROUP BY OrderStatus  
ORDER BY Order_Count DESC;
```



MySQL Workbench interface showing the execution of a SQL query. The query is a GROUP BY statement on OrderStatus, ordered by Order_Count in descending order. The results show 5 rows: Returned (145), Cancelled (143), Delivered (142), Shipped (140), and Pending (120). The bottom panel shows the execution log with 8 messages.

OrderStatus	Order_Count	Gross_Value	Avg_Order_Value
Returned	145	134725.08	929.14
Cancelled	143	162853.32	1138.83
Delivered	142	142758.88	1005.34
Shipped	140	130322.67	930.88
Pending	120	141458.77	1178.82

Execution Log:

#	Time	Action	Message	Duration / Fetch
27	09:26:46	SELECT OrderID, Date, CustomerID, Product, Quantity, UnitPrice, TotalPrice, OrderStatus FROM ecom_orders WHERE OrderStatus = 'Delivered' A...	51 rows(s) returned	0.016 sec / 0.000 sec
28	09:27:17	SELECT OrderID, Date, CustomerID, Product, Quantity, UnitPrice, TotalPrice, OrderStatus FROM ecom_orders WHERE OrderStatus = 'Delivered' A...	51 rows(s) returned	0.000 sec / 0.000 sec
29	09:27:41	SELECT OrderStatus, COUNT(OrderID) AS Order_Count, ROUND(SUM(TotalPrice), 2) AS Gross_Value, ROUND(AVG(TotalPrice), 2) AS ...	5 rows(s) returned	0.016 sec / 0.000 sec
30	09:31:21	SELECT OrderID, Date, CustomerID, Product, Quantity, UnitPrice, TotalPrice, OrderStatus FROM ecom_orders WHERE OrderStatus = 'Delivered' A...	51 rows(s) returned	0.000 sec / 0.000 sec
31	09:47:49	SELECT Product, COUNT(OrderID) AS Total_Orders, SUM(Quantity) AS Total_Units_Sold, ROUND(AVG(UnitPrice), 2) AS Avg_Unit_Pric...	7 rows(s) returned	0.016 sec / 0.000 sec
32	09:49:20	SELECT OrderStatus, COUNT(OrderID) AS Order_Count, ROUND(SUM(TotalPrice), 2) AS Gross_Value, ROUND(AVG(TotalPrice), 2) AS ...	5 rows(s) returned	0.000 sec / 0.000 sec

Key Finding: Returned orders were the highest volume post-2024 (145 orders), followed by Delivered and Shipped.

5. Query 4: Marketing Referral Channels

Objective

Calculate fulfilled orders, units sold, and net revenue by referral source for Shipped and Delivered orders.

SQL

SELECT

```
ReferralSource,  
COUNT(OrderID) AS Fulfilled_Orders,  
SUM(Quantity) AS Total_Units,  
ROUND(SUM(TotalPrice), 2) AS Net_Revenue  
FROM ecom_orders  
WHERE OrderStatus IN ('Shipped', 'Delivered')  
GROUP BY ReferralSource  
ORDER BY Net_Revenue DESC;
```

The screenshot displays the MySQL Workbench interface. The central SQL editor contains the following query:

```
4 COUNT(OrderID) AS Fulfilled_Orders,  
5 SUM(Quantity) AS Total_Units,  
6 ROUND(SUM(TotalPrice), 2) AS Net_Revenue  
7 FROM ecom_orders  
8 WHERE OrderStatus IN ('Shipped', 'Delivered')  
9 GROUP BY ReferralSource  
10 ORDER BY Net_Revenue DESC;
```

Below the editor, the 'Results' tab shows a table with the following data:

ReferralSource	Fulfilled_Orders	Total_Units	Net_Revenue
Instagram	94	284	108246.78
Email	96	286	100147.67
Google	93	277	95367.23
Facebook	91	262	93189.6
Referral	92	261	91608.62

The bottom panel shows the 'Output' tab with a log of SQL execution actions, including timestamps and messages for each query step.

Key Finding: Instagram generated highest net revenue (\$108,246.78 across 94 orders), followed by Email (\$100,147.67).

6. Summary

All four SQL queries were successfully executed, verified, and analyzed against the dataset to extract actionable insights on revenue performance, order fulfillment status, and marketing referral effectiveness.